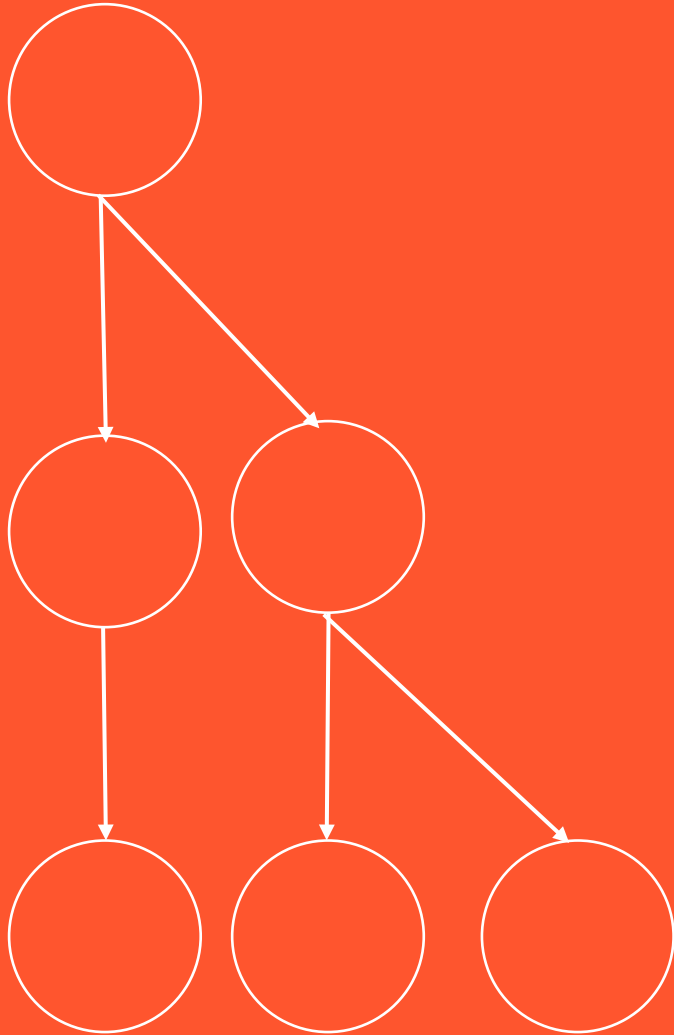


BCOG 100

Pattern Recognition

Lecture 2: Pattern Recognition Algorithms

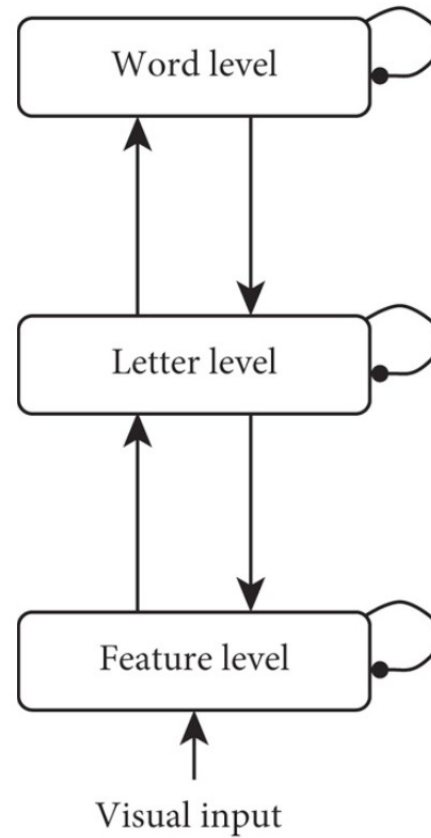


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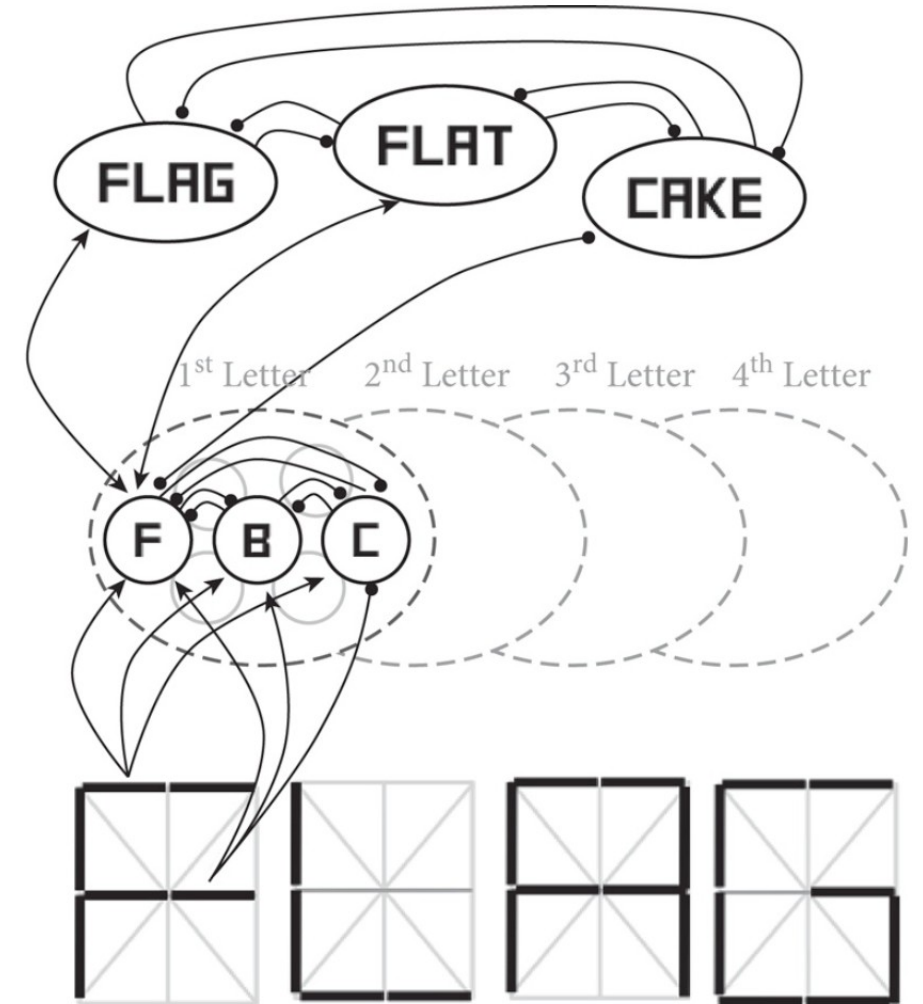
Reading Check 7-2

Password: ia

Question: Explain
this figure



Click to view larger

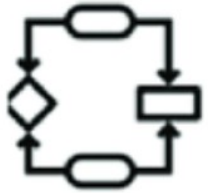


Pattern Recognition Algorithms



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

The problems of pattern recognition

Today:

- How systems overcome these problems
- What principles make recognition adaptive and robust

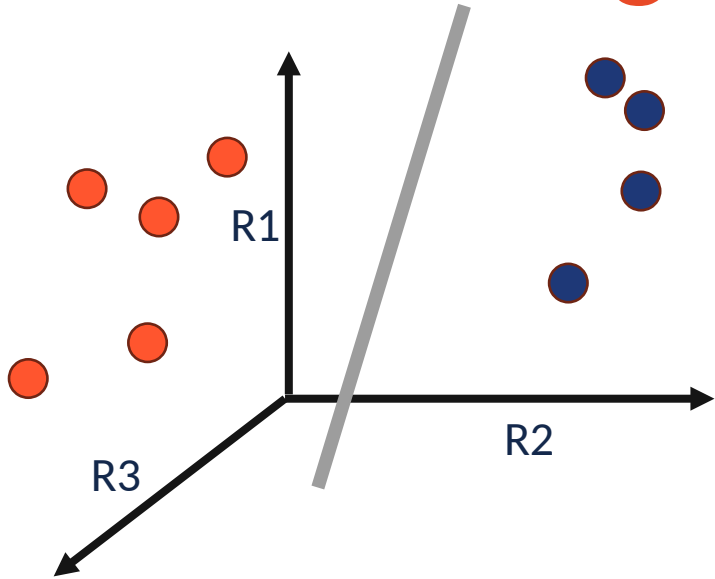
Pattern Recognition Algorithms



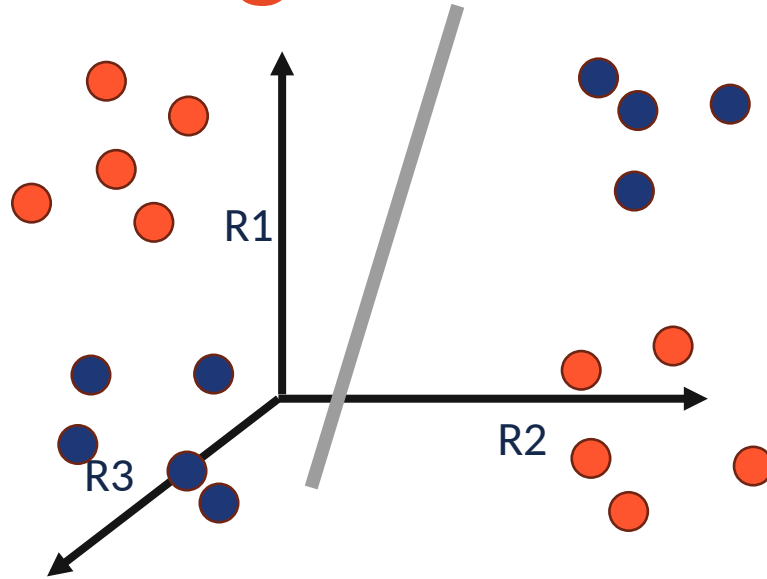
From Limits to Learning

- Brains, behavior, and AI face similar challenges
- Evolution, psychology, and ML found convergent solutions
- Representation, abstraction, and learning over time

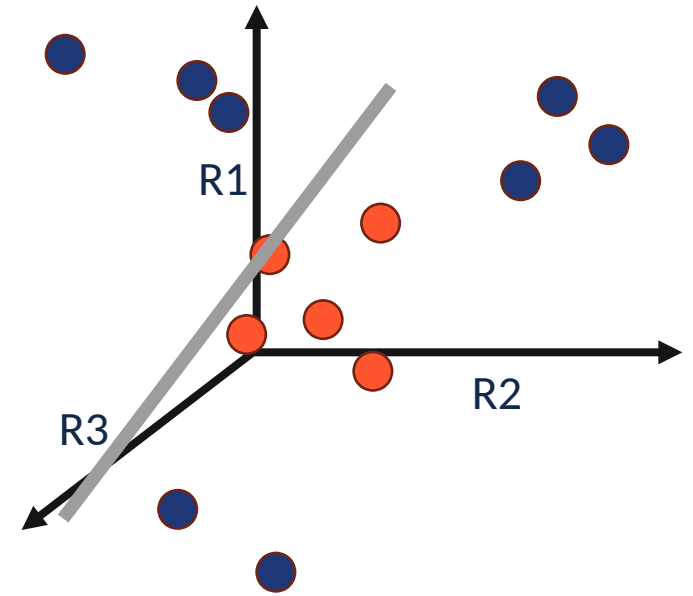
Pattern Recognition Algorithms



Linearly Separable



XOR

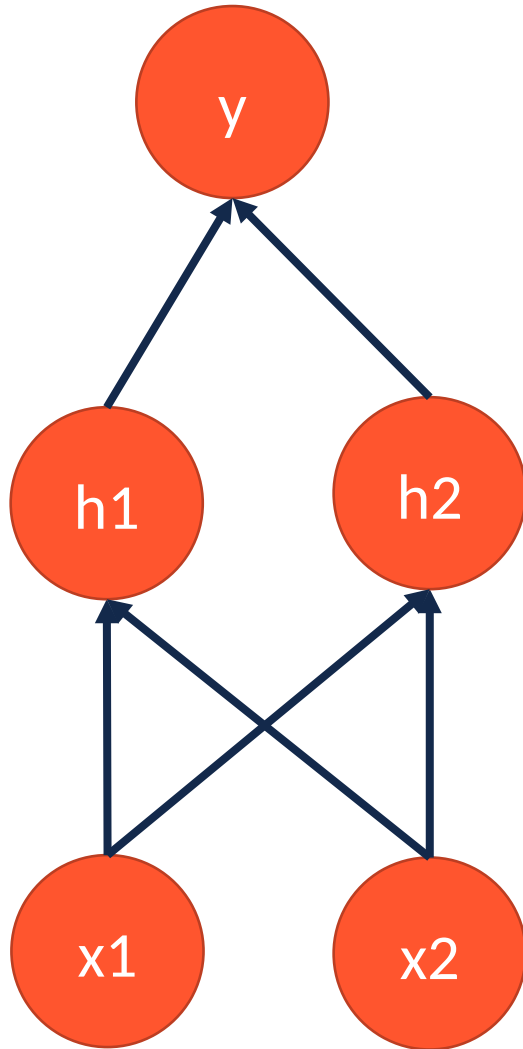


Prototype

Linear Boundaries and Their Limits

- Single-layer systems draw one boundary
- Good for simple categories, not complex ones
- Example: The exclusive OR problem

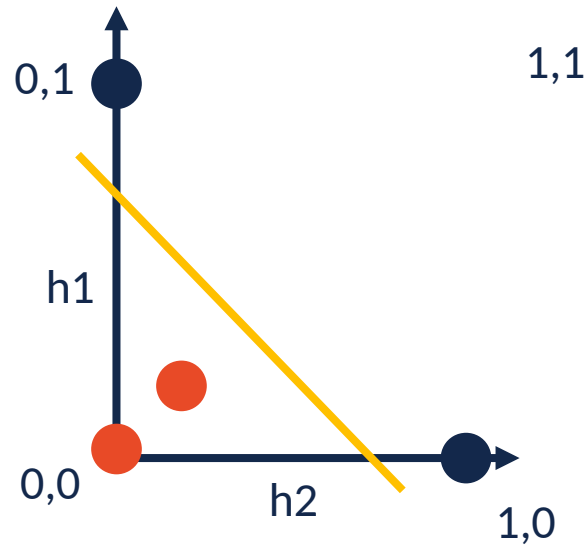
Pattern Recognition Algorithms



Nonlinear Solutions: Hidden Layers

- Intermediate layers transform the feature space
- Nonlinearity enables flexible separation
- Similar logic appears in sensory systems

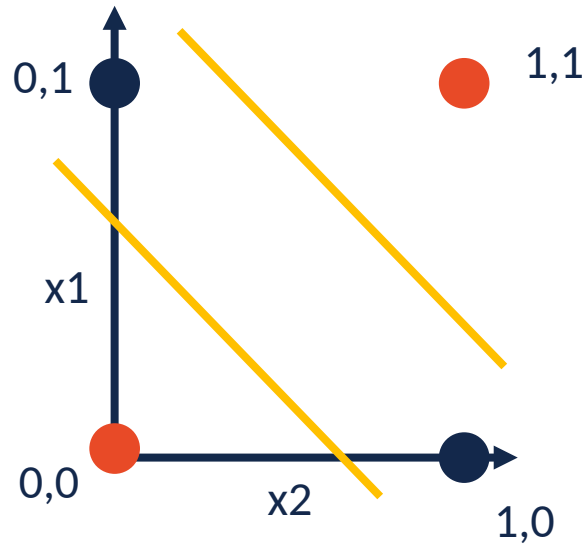
Pattern Recognition Algorithms



Hidden Layers as Geometry

- Remapping inputs into a new latent space
- Warping feature space until categories become separable
- Hidden layers as learned coordinate transformations

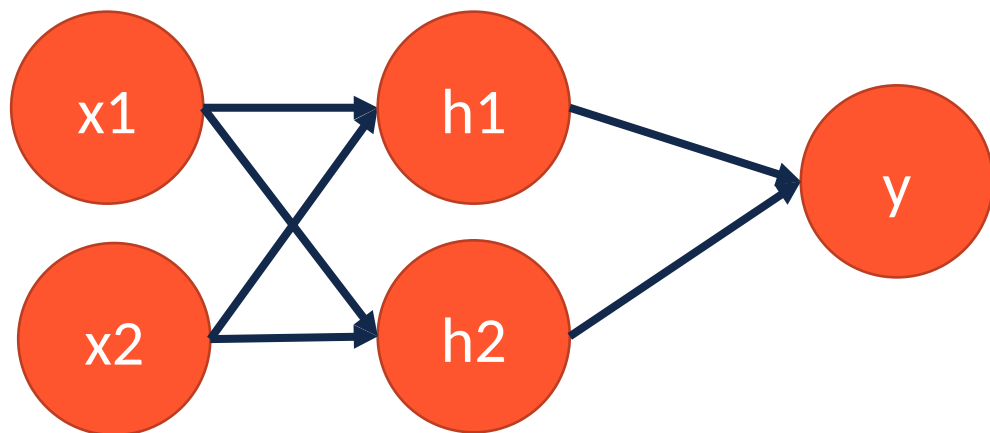
Pattern Recognition Algorithms



Hidden Layers as Decision Boundaries

- Each unit draws one linear cut in space
- The output combines them into curved or disjoint regions
- Complex boundaries built from many simple ones

Pattern Recognition Algorithms



XOR		
x1	x2	y
0	0	0
0	1	1
1	0	1
1	1	0

NOT AND		
x1	x2	h1
0	0	1
0	1	1
1	0	1
1	1	0

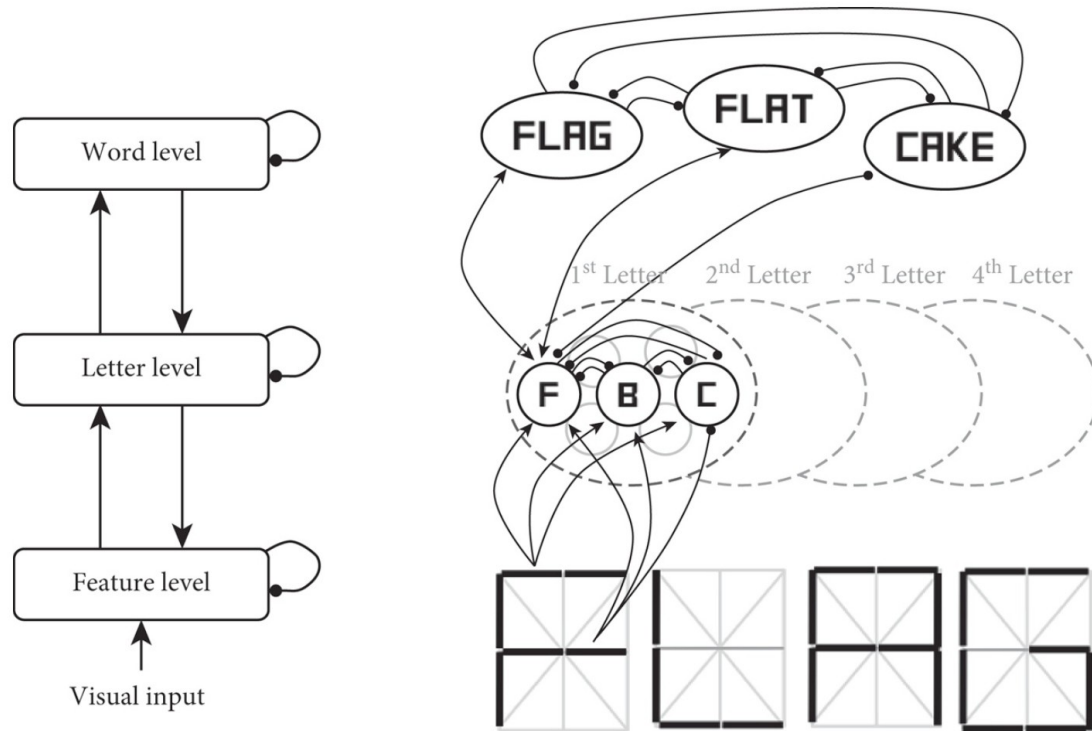
OR		
x1	x2	h2
0	0	0
0	1	1
1	0	1
1	1	1

AND		
h1	h2	y
1	0	0
1	1	1
1	1	1
0	1	0

Hidden Layers as Compositional Logic

- Each unit performs a simple logical function
- Combining layers composes clauses into complex rules
- Networks implement compositional Boolean structure

Pattern Recognition Algorithms



Click to view larger

Connectionism in Cognitive Psychology

- PDP models formalized learning as weighted networks
- Paralleled psychological theories of perception and memory
- Cognitive science anticipated deep learning principles

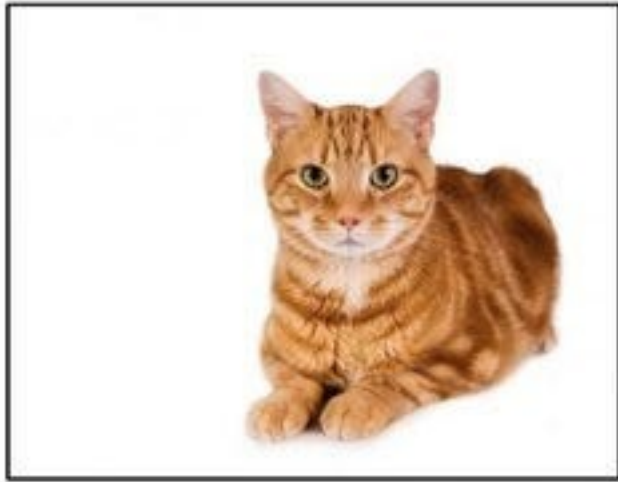
Pattern Recognition Algorithms

		Localist Representation						Distributed Representation					
		Bear	Tiger	Eagle	Fish	Turtle	Frog	Walk	Swim	Fly	Egg	Claw	Wild
Bear		●						●	●	○	○	●	●
Tiger			●					●	●	○	○	●	●
Eagle				●				○	○	●	●	●	○
Gold Fish					●			○	●	○	●	○	○
Turtle						●		●	●	○	●	○	○
Frog							●	●	○	○	●	●	○

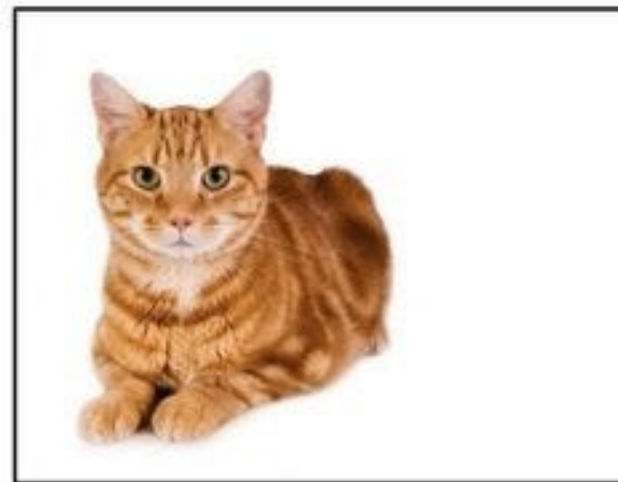
Localist vs. Distributed Representations

- Localist codes are explicit and interpretable
- Distributed codes are flexible and generalizable
- Brains and networks often combine both

Pattern Recognition Algorithms



Cat



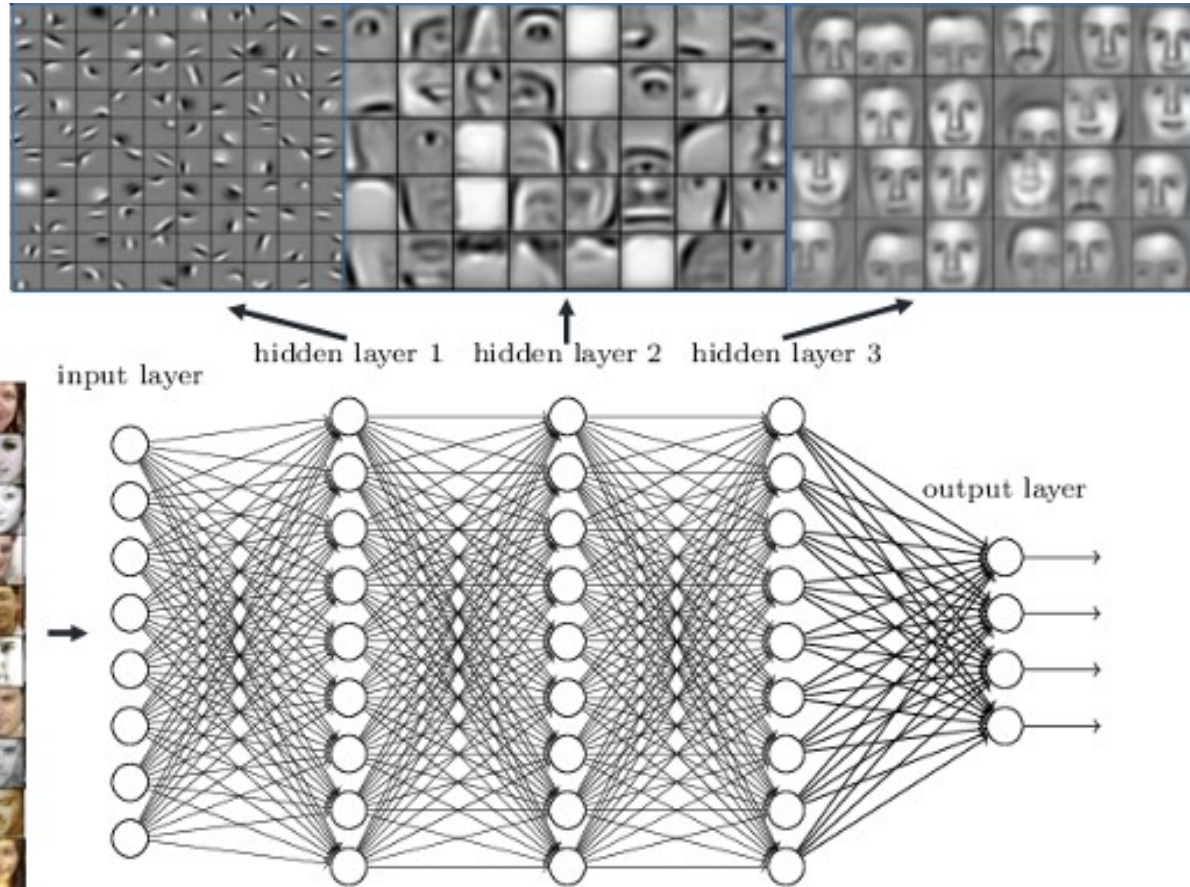
Cat

Hierarchical Representations and Abstraction

- Recognizing the same thing in different forms
- Layers capture structure from local to global
- Composition enables invariance and generalization

Pattern Recognition Algorithms

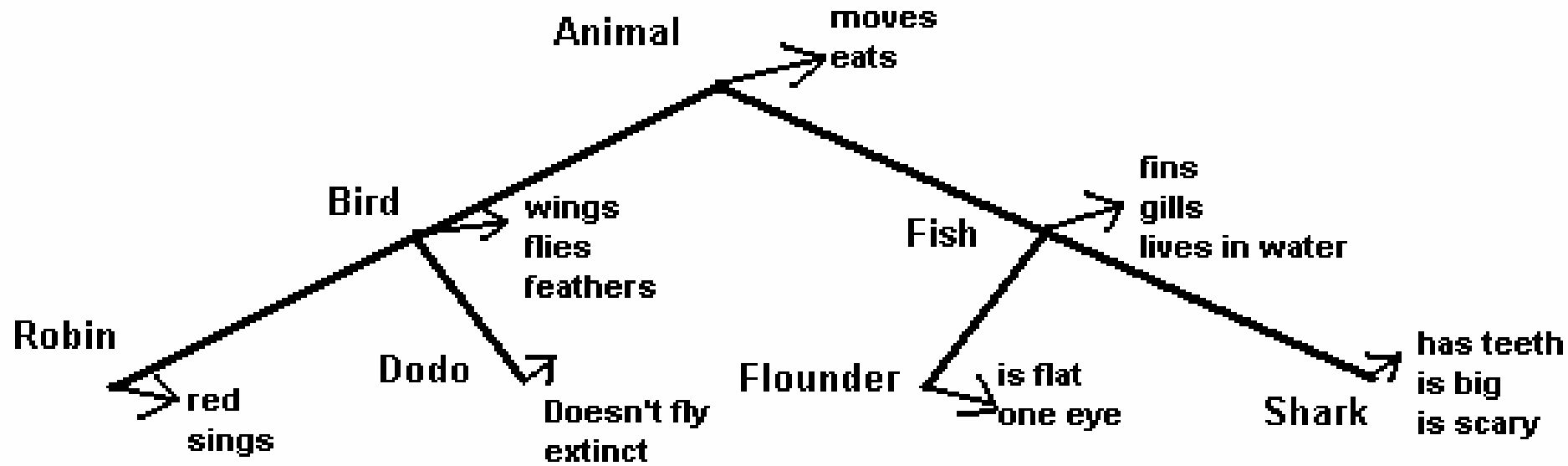
Deep neural networks learn hierarchical feature representations



Convolutional Neural Networks (CNNs)

- Networks that learn visual hierarchies automatically
- Detect local features, build global representations
- Inspired by—and similar to—the visual cortex

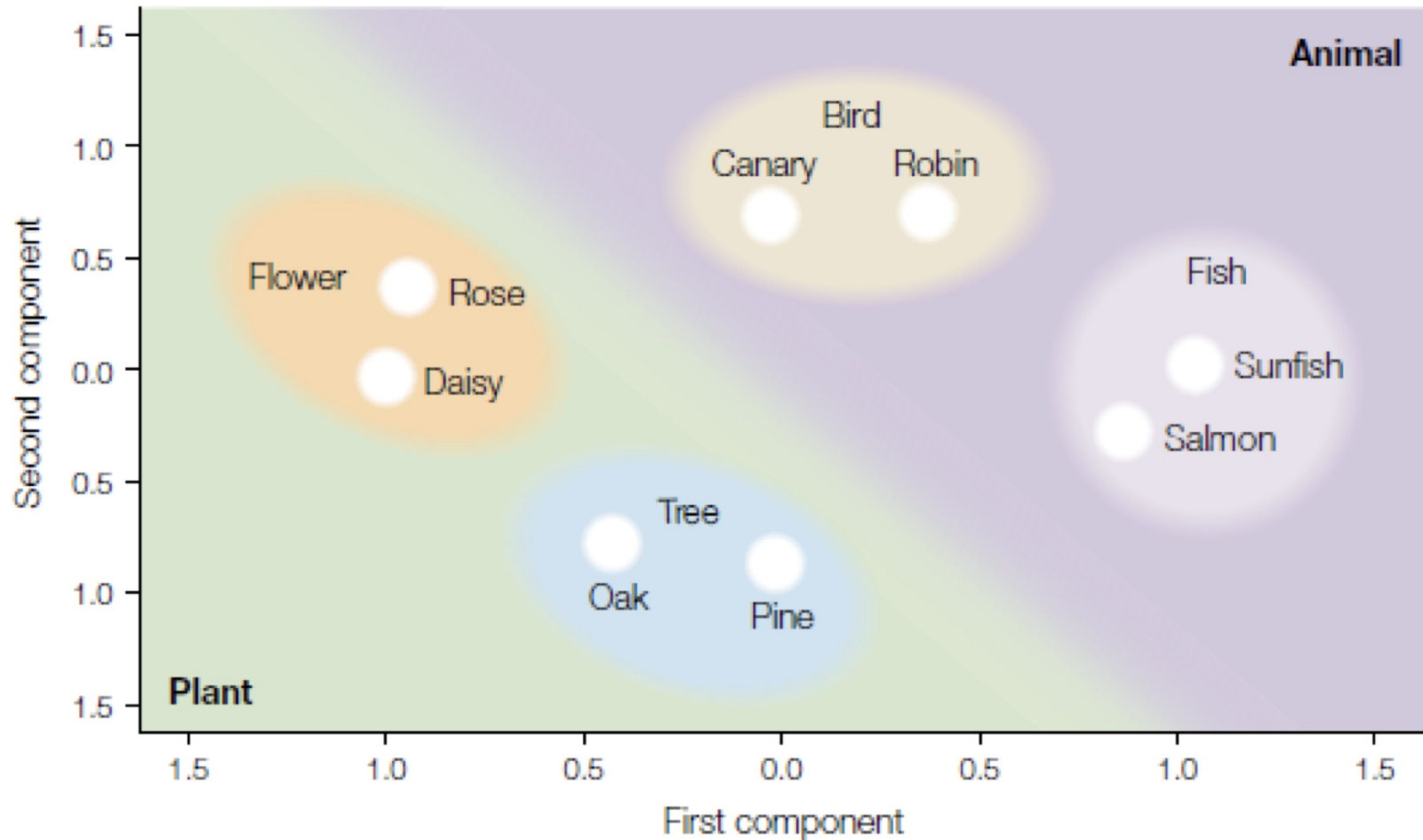
Pattern Recognition Algorithms



Hierarchies in Psychology

- Collins and Quillian's semantic hierarchies
- Biederman's recognition-by-components
- Chomsky's hierarchical syntax in language

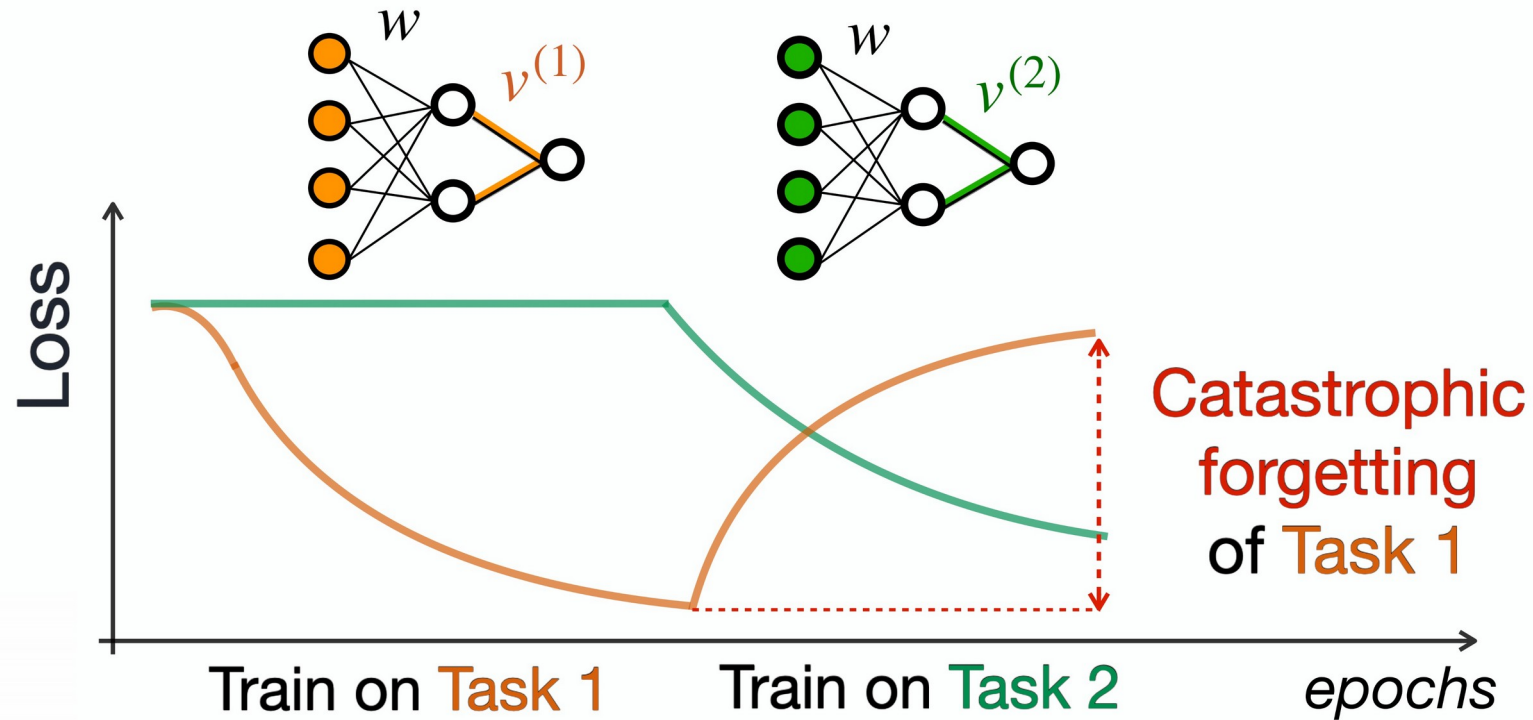
Pattern Recognition Algorithms



What Hierarchies Solve —and Don't

- Neural networks learn hierarchies implicitly, not explicitly
- Implicit hierarchies support flexible generalization
- But lack explicit structure for relations and meaning

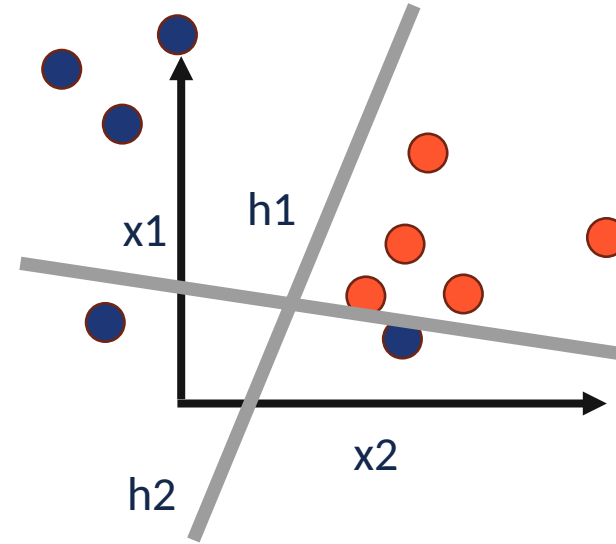
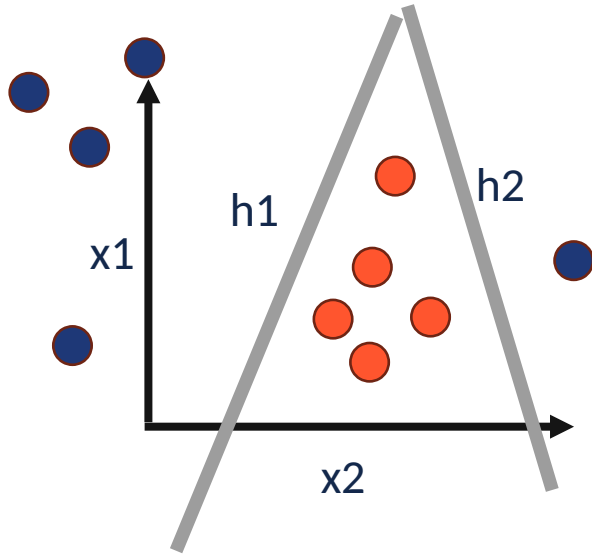
Pattern Recognition Algorithms



Continual Learning: Stability vs. Plasticity

- Learning without forgetting is difficult
- Catastrophic interference in neural networks
- Arithmetic analogy: learning subtraction disrupts addition

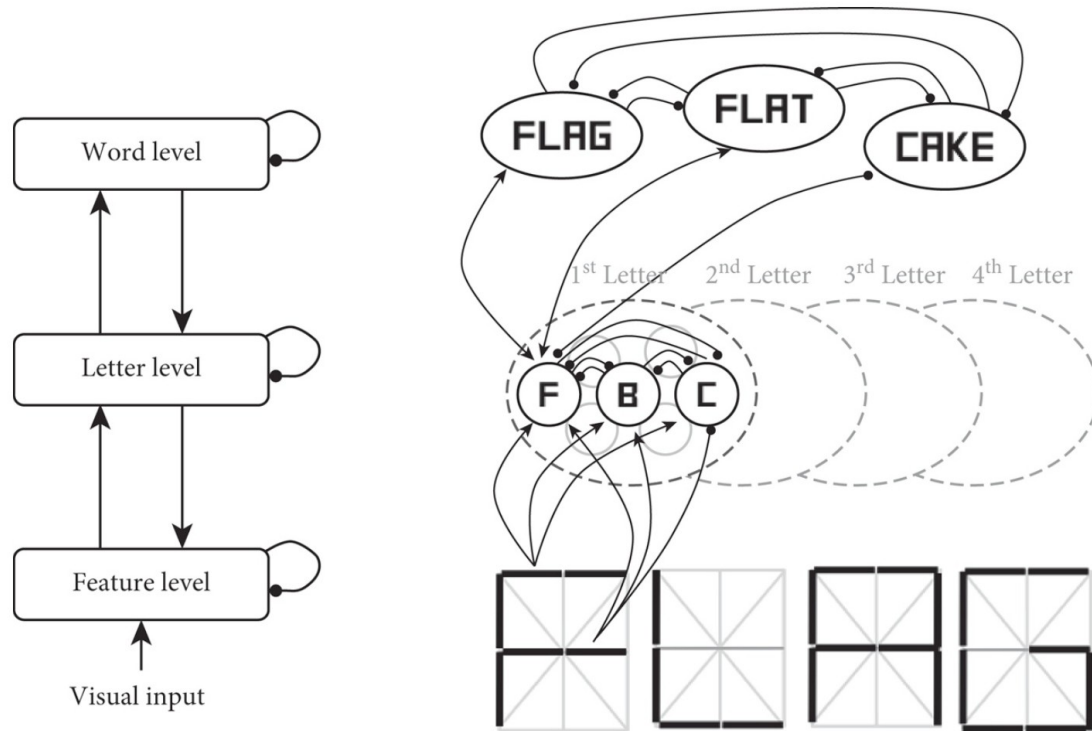
Pattern Recognition Algorithms



Why Forgetting Happens

- Distributed weights store overlapping knowledge
- New learning reshapes existing representations
- Brains mitigate this through consolidation and replay

Pattern Recognition Algorithms



Click to view larger

Strengths of Modern Neural Networks

- Learn complex nonlinear mappings
- Discover hierarchical and distributed structure
- Perform impressive pattern recognition tasks

Pattern Recognition Algorithms

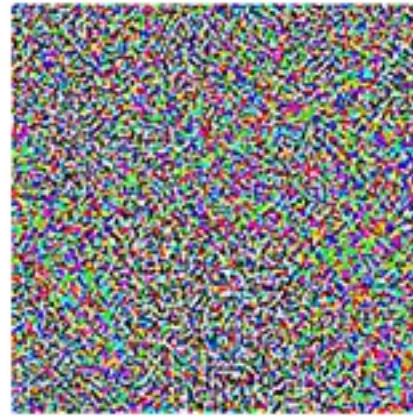


x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



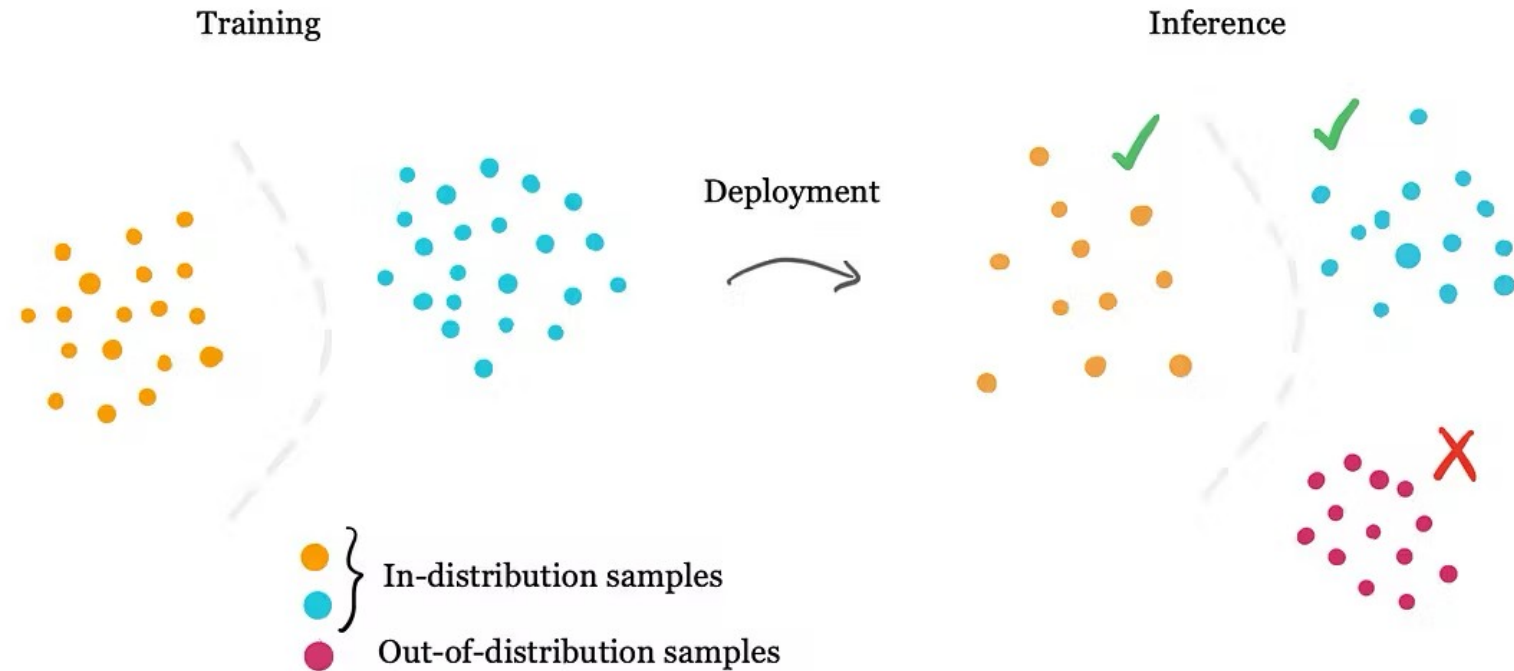
$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$
“gibbon”

99.3 % confidence

Adversarial Fragility

- Tiny pixel changes cause confident misclassification
- Dependence on shallow statistical cues
- Humans remain robust to these distortions

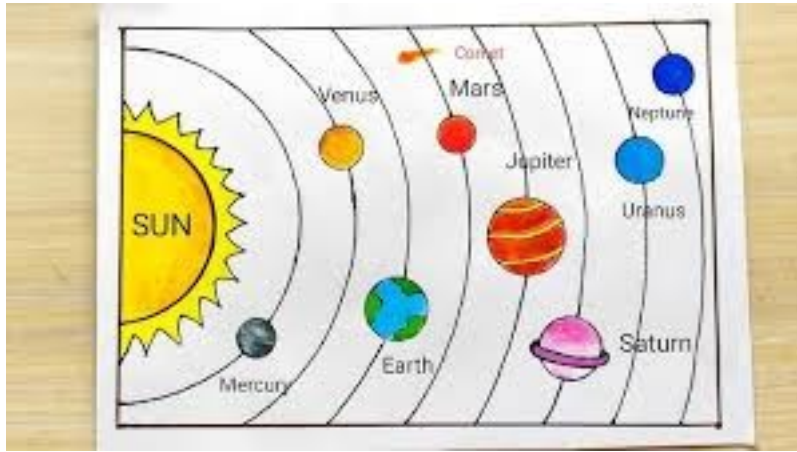
Pattern Recognition Algorithms



Out-of-Distribution Generalization

- Performance collapses outside training conditions
- Strong interpolation, weak extrapolation
- Contrast with human flexibility across contexts

Pattern Recognition Algorithms

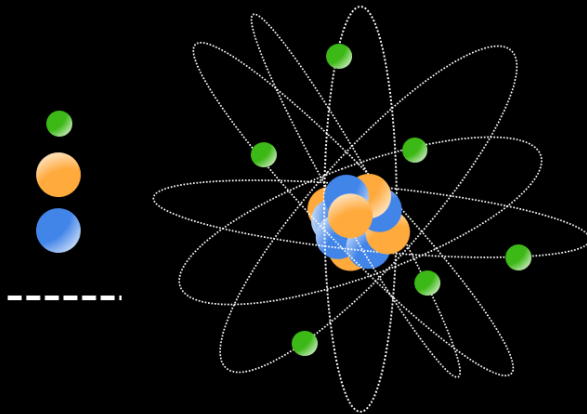


Relational and Causal Reasoning

- Networks capture correlation, not relation or cause
- Humans infer structure and intervene on the world
- Embodied experience may be key to this gap

Simple Atomic Structure

Electrons
Protons
Neutrons
Electron Orbit



Pattern Recognition Algorithms



Humans and Animals vs. ANNs

- Fish, mice, and bees show robust recognition
- Networks match some abilities but remain brittle
- These gaps reveal both limits and insights

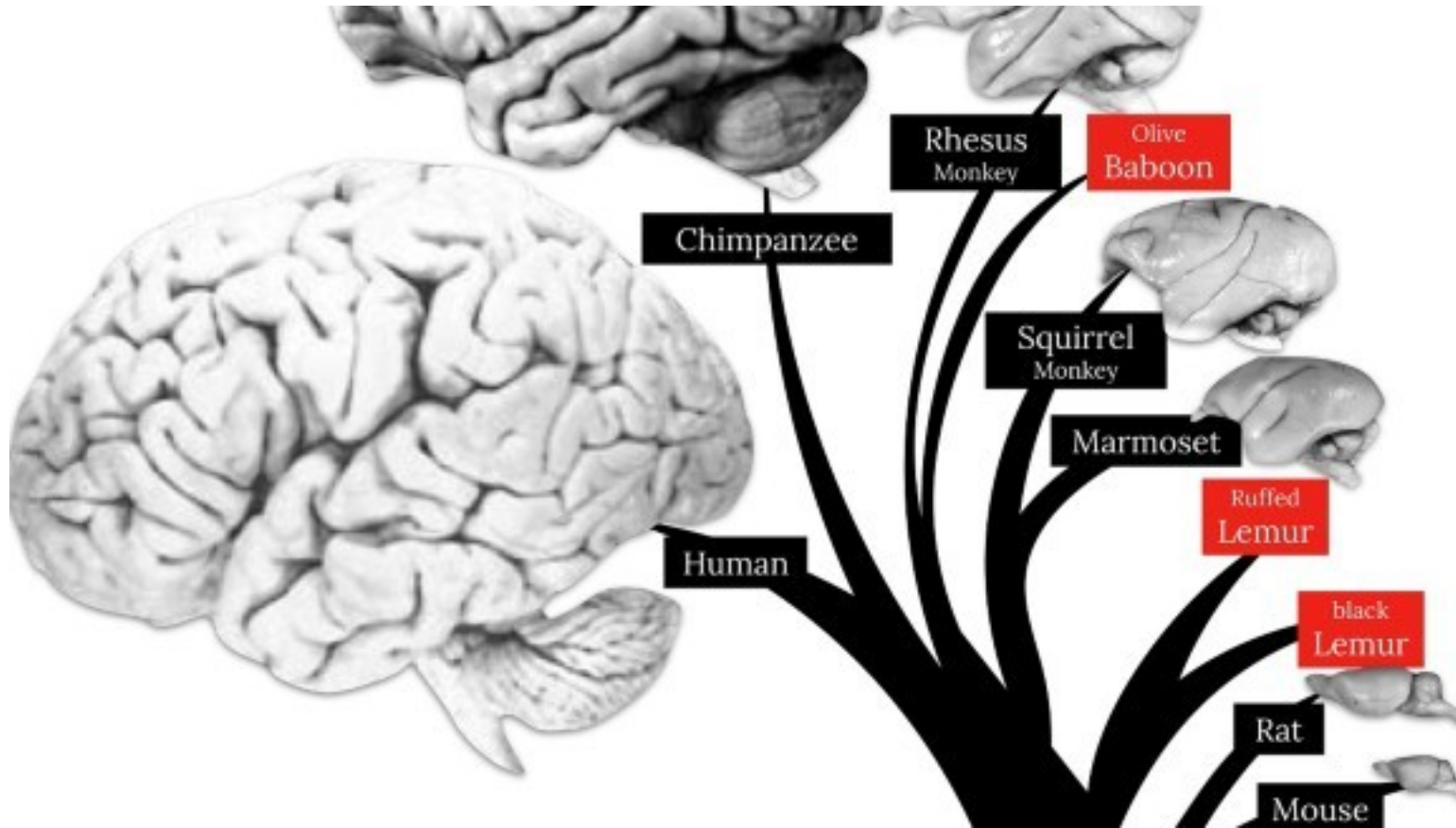
Pattern Recognition Algorithms



Two Kinds of Missing Pieces

- Missing fundamentals: embodiment, recurrence, integration, explicit structure
- Later-evolving systems: causal, analogical, symbolic
- Pattern recognition is only the foundation

Pattern Recognition Algorithms



Lessons from Evolution

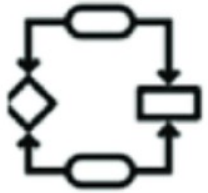
- Early perceptual intelligence appears in both animals and ANNs
- Later systems build on these foundations
- Progress mirrors evolutionary layering

Pattern Recognition Algorithms



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap

- Neural networks solve many (but not all) of the problems a pattern recognition system needs to solve

Up Next

- How brains implement feature extraction and learning
- Neural architecture of stability and hierarchy